

5057/6

FACULTY OF COMMERCE & BUSINESS MANAGEMENT
B.Com. (V-Semester) Examination
(Computer Applications)
Paper-III (New)
Relational Database Management Systems

Time: 3 Hours]

[Max. Marks: 50

Section – A (Marks: 5 x 3 = 15)

1 Answer any **five** of the following Questions

- (a) What is DBMS?
- (b) Key Constraint and Integrity Constraint.
- (c) What is Normalization?
- (d) Hashed File Organization.
- (e) What DML commands?
- (f) What are Joins?
- (g) What is Transaction?
- (h) What is OLAP?

Section – B (Marks: 5 x 7 = 35)
Answer all the **following** question.

- 2 (a) Explain DBMS three-level architecture.
Or
(b) Explain the process of creating an Entity-Relationship Diagram for a college database.
 - 3 (a) What are the different types of keys used in database? Give examples.
Or
(b) What is Index? Explore different types of indices.
 - 4 (a) Explain the different data types used in SQL with examples.
Or
(b) What is View? Explain different operations performed on View with example.
 - 5 (a) What is Lock? Explain the types of locking techniques.
Or
(b) Explain database security.
 - 6 (a) Explain the advantages and disadvantages of DDBMS.
Or
(b) Describe about Data Fragmentation in detail.
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FACULTY OF SCIENCE
B.Sc.(I Semester) Examination
(Computer Science/Computer Applications)
Paper I
Programming in C

[Max Marks:80]

Time:3 Hours]

Section A (Marks: 8x4=32)

1 Answer any eight questions:

- a) Differentiate RAM and ROM?
- b) Define Algorithm and write Algorithm to find the largest of three numbers
- c) Discuss in short about increment and decrement operator
- d) Explain simple if () with example program.
- e) Explain do—while with suitable example program.
- f) Explain about c type. hand few character built in functions
- g) Explain about Scope of a variable
- h) Write a program for Linear search.
- i) Write a short note on pointers.
- j) Discuss about union and its declaration
- k) Explain about Enumeration type
- l) What is a file? Discuss modes of opening a file.

Section B (Marks: 4x12=48)

Answer **all** questions

2 a) Explain briefly generations and classification of programming languages
OR

b) Write Pseudo code and draw the flow chart to check the given number is prime or not.

3 a) Differentiate Switch case with else if ladder with suitable example program.
OR

b) Discuss usage of Break and Continue Statements with suitable example programs

4 a) Discuss in detail the String Manipulation functions
OR

b) Discuss in detail various valid arithmetic operations that can be performed on pointers in C with example program

5 a) Define structure? Explain arrays of structures in detail with suitable example program.

OR

b) Discuss about different operations performed on file? Write about Random Access files?

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FACULTY OF COMMERCE & BUSINESS MANAGEMENT
B.Com. (I Semester) Examination
(Computer Applications)
Paper – III (New)
FUNDAMENTALS OF INFORMATION TECHNOLOGY

Time: 3 Hours]

[Max. Marks: 50

Section – A (Marks: 5 x 3 = 15)

1. Answer any FIVE Questions:

- a) System and Application Software
- b) Cloud Vs. Local Storage
- c) VLOOKUP
- d) Routers Vs. Switches
- e) Star Topology
- f) DBMS
- g) Client-Server Architecture
- h) Safe browsing practices

Section – B (Marks 5 x 7 = 35 Marks)

Answer all the Questions

2 a) Describe the components of Computer System.

(Or)

b) Examine the strategic role of IT in functional areas of business units.

3 a) Discuss the different types of Storage devices.

(Or)

b) Elucidate the role of data in business decision-making.

4 a) Analyze the functions and types of Operating Systems.

(Or)

b) Discuss the features of Ms-Word.

5 a) Define Network. Discuss the different types of Networks.

(Or)

b) Discuss the various E-Commerce models.

6 a) Explain the characteristics and advantages of Cloud Computing.

(Or)

b) Discuss the different types of Cyber Security threats.

FACULTY OF SCIENCE
B.Sc. (I-Semester) Examination
COMPUTER SCIENCE
Paper-I (New)
Programming in C

[Max Marks: 80]

Time: 3 Hours]

Section-A (Marks: 8 x 4 =32)

1 Answer any **Eight** of the following:

- a) Classify programming languages by generation and give example of each.
- b) Differentiate between compilation and interpretation.
- c) List the C tokens and give one example of each.
- d) State the escape sequences and their usage in input/output operations.
- e) Define while loop, syntax and illustrate with an example.
- f) What is one-dimensional array? Show its declaration and initialization.
- g) Define a function in C. Write its definition, declaration, and calling syntax.
- h) What is recursion? Write a recursive function to calculate factorial.
- i) Write the scope and lifetime of local and global variables.
- j) What is structure in C and show how to declare and initialize it?
- k) Define an enumeration (enum) type, and give an example declaration.
- l) Write the syntax of fopen() and fclose(), and state their purpose.

Section B - (Marks 4 x 12 =48)

Answer all questions

- 2 a) Describe the four processes – compilation, interpretation, loading, and linking – and state the role of each.

Or

- b) List arithmetic, relational, and logical operators in C. Evaluate the expression $Sa = 5, b = 3, c = 2; \text{result} = a + b * c - a / b;$ showing operator precedence and associativity.

- 3 a) Explain un-formatted (getchar, putchar, gets, puts) and formatted (scanf, printf) I/O functions with syntax and one example each.

Or

- b) Briefly explain goto, break, continue, return, and exit statements with one example each.

- 4 a) Describe call-by-value and call-by-reference with syntax and one example program each.

Or

- b) Describe malloc, calloc, and free functions – their syntax, usage, and write a program to allocate an integer array dynamically.

- 5 a) Define Array of Structure. How it can be declared? Explain with an example.

Or

- b) Give an brief overview of file handling in C. List the steps for opening, processing, and closing a file.

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FACULTY OF SCIENCE
B.Sc. (II-Semester) Examination
COMPUTER SCIENCE
Paper-II
Programming in C++

[Max. Marks: 80]

Time: 3 Hours]

Section – A (Marks: 8 x 4 = 32)

1 Attempt any **Eight** of the following questions:

- a) Write about Arrays.
- b) Write a short note on Inline Functions.
- c) Write about Benefits of OOPs.
- d) What are destructors? When they are called? What is their utility?
- e) Write about static keyword.
- f) How to create classes and objects in C++? Give example.
- g) Write the significance of pure virtual functions in C++.
- h) Give a brief note on Abstract Base Class.
- i) Explain Stream classes hierarchy with suitable example.
- j) Write about the Objects of the class Template.
- k) Discuss the benefits of exception handling.
- l) What is rethrowing an exception?

Section – B (Marks: 4 x 12 = 48)

Answer All Questions

- 2 a) Discuss about User defined functions by passing data by Value and Reference with suitable example programs.

(OR)

- b) Discuss various Branching Statements in C++ With syntaxes and suitable examples.

- 3 a) What is Constructor? Explain about different types of constructors with suitable Program example.

(OR)

- b) Discuss in detail about operator overloading with suitable examples and write a program for Unary plus operator overloading.

- 4 a) Discuss about the different types of Inheritances with suitable example codes.

(OR)

- b) What is polymorphism in C++? Explain about its types with suitable examples.

- 5 a) Discuss in detail about the different types of exceptional handling constructs in c++ with suitable code program example.

(OR)

- b) Explain the Overloading with Function Templates with suitable code.

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FACULTY OF SCIENCE
B.Sc. (II-Semester) Examination
COMPUTER SCIENCE

Paper-II
Programming in C++

Time: 3 Hours]

[Max. Marks: 80

Section – A (Marks: 8 x 4 = 32)

1 Attempt any **Eight** of the following questions:

- a) Write about Arrays.
- b) Write a short note on Inline Functions.
- c) Write about Benefits of OOPs.
- d) What are destructors? When they are called? What is their utility?
- e) Write about static keyword.
- f) How to create classes and objects in C++? Give example.
- g) Write the significance of pure virtual functions in C++.
- h) Give a brief note on Abstract Base Class.
- i) Explain Stream classes hierarchy with suitable example.
- j) Write about the Objects of the class Template.
- k) Discuss the benefits of exception handling.
- l) What is rethrowing an exception?

Section – B (Marks: 4 x 12 = 48)

Answer All Questions

- 2 a) Discuss about User defined functions by passing data by Value and Reference with suitable example programs.

(OR)

- b) Discuss various Branching Statements in C++ With syntaxes and suitable examples.

- 3 a) What is Constructor? Explain about different types of constructors with suitable Program example.

(OR)

- b) Discuss in detail about operator overloading with suitable examples and write a program for Unary plus operator overloading.

- 4 a) Discuss about the different types of Inheritances with suitable example codes.

(OR)

- b) What is polymorphism in C++? Explain about its types with suitable examples.

- 5 a) Discuss in detail about the different types of exceptional handling constructs in c++ with suitable code program example.

(OR)

- b) Explain the Overloading with Function Templates with suitable code.

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FACULTY OF SCIENCE

B.Sc (IV- Semester) Examination

COMPUTER SCIENCE

Paper-IV

Database Management Systems

Time:3 Hours]

[Max.Marks:80

Section-A (Marks: 8 x 4 = 32)

1 Answer any **Eight** of the following:

- a) List the functions of Database Administrator.
- ~~b)~~ Write about different types of Keys.
- c) Describe the fundamental operations in relational algebra.
- d) What do you mean by scheme refinement? Explain how it can be accomplished?
- e) Explain what the problems are caused by redundancy.
- f) Give an example of a relation scheme R and a set of dependencies such that R is in BCNF but not in 4 NF?
- g) Explain the following Operators in SQL with examples: i) SOME ii) IN.
- ~~h)~~ Explain with an example about aggregate operators.
- i) Write about Views in SQL.
- ~~j)~~ What is serializability?
- k) Explain Failure with loss of nonvolatile storage.
- ~~l)~~ Explain the ACID Properties of transactions.

Section-B (Marks: 4x12=48)

Answer all questions

2 a) Give an overview of database architecture.

Or

- b) i) Write Relational Algebra Queries for the following (for Sailors Database)
Sailors(sid,sname, rating, age), Boats(bid, bname, color), Reserves(sid, bid, day).
(I) Find the Sailorid's with age over 20 and who have not reserved a red boat.
(II) Find the names of Sailors who reserved boat 103 .
ii) Explain tuple relational calculus.

3 a) Describe the components of an ER diagram. Construct an ER-Diagram for a hospital with a Set of patients and set of medical doctors associated with each patient a log of the various tests and examinations conducted.

Or

- ~~b)~~ Explain 2NF, 3NF and BCNF Normal forms with example. What is the difference between 3NF and BCNF?

[P.T.O

-2-

- 4 a) Describe various data types in SQL. Consider the following relations: Hotel (Hotel_no, Hotel_name, City) Room (Room_no, Hotel_no, Type, Price) Booking (Hotel_no, Guest_no, DateFrom, DateTo, Room_no) Guest (Guest_no, GuestName, GuestAddress) Write the appropriate queries in SQL for the following: i) Find the average price of a room ii) List the names and address of all guests with bookings for a hotel in London, alphabetically ordered by name iii) Find the total income from all the rooms of the hotels in NewYork iv) List the Name(s) of Guest(s) at the winner hotel, who are paying highest price for a room.
- Or
- b) List out various Integrity Constraints in SQL. Also define trigger and Explain how to implement triggers in SQL?
- 5 a) What is transaction? Give an overview of validation based protocol.
- Or
- b) Explain Deadlock Prevention and Recovery mechanisms.
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FACULTY OF SCIENCE
B.Sc. (III-Semester) Examination
COMPUTER SCIENCE
Paper-III
Data Structures Using C++

[Max Marks: 80]

Time: 3 Hours]

Section-A (Marks: 8 x 4 =32)

1 Answer any **Eight** of the following:

- a) Convert the $\frac{(x-y*c)}{a} + (z-b)/(d+e)$ expression to postfix and prefix
- b) Define Algorithm and Pseudo code? Differentiate them.
- c) Discuss about processing of a function calls.
- ~~d)~~ Explain about Priority Queue.
- e) Discuss about the difference between Recurrence and Recursion with example.
- ~~f)~~ Write Algorithm for Linear Search.
- ~~g)~~ Write a short note on AVL Tree.
- h) Define Symbol Table. What are the uses of it?
- ~~i)~~ Illustrate the Bubble sort with example.
- ~~j)~~ Write about Spanning Tree.
- k) Give a brief note on Hash Functions.
- l) Discuss about extendable hashing mechanism in detail.

Section B - (Marks 4 x 12 =48)

Answer all questions

- 2 a) Explain how infix expressions are converted to polish notation. Illustrate your answer with suitable example.
Or
~~b)~~ Discuss about the representation of Stack Using Arrays and operations of Stack.
- 3 a) What is doubly linked list? Write the algorithms for insertion and deletion of an element in to doubly linked list.
Or
b) What is Sparse Matrix? Discuss how it is represented by using Linked List.
- 4 a) Define Binary Tree. Write the Pseudo Code for insertion and deletion operations on Binary Tree.
Or
~~b)~~ Illustrate about the Merge Sort and Quick Sort in detail.
- 5 a) Illustrate the Kruskal's Algorithm with suitable example.
Or
~~b)~~ Discuss about Graph Tree Traversal mechanisms in detail.

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FACULTY OF SCIENCE
B.Sc. (III SEMESTER) EXAMINATION
COMPUTER SCIENCE
Paper III
DATA STRUCTURES Using C++

Time: 3 Hours]

[Max.Marks:80

SECTION-A (Marks: 8X4=32)

1 Answer any **Eight** Questions

- a) What is Pseudo code? Explain with an example.
- b) Convert the $(x - b) * y - \frac{a}{c} - (d * f - e)$ expression to postfix and prefix.
- c) Write a short note on Abstract Data type
- d) Give a brief note on Analysis of Algorithms
- e) Write about the Representation of Sparse Matrix using Linked List
- f) Write the differences between Iteration versus Recursion
- g) Illustrate the bubble sort Algorithm with suitable example
- h) Write a short note on types of Trees
- i) Write about Symbol Tree
- j) Explain about Hash Table Overflow.
- k) Define directed Graph
- l) Write about Heap applications

SECTION-B (Marks: 4X12=48)

Answer all questions

- 2 a) What is Pseudo code? Explain with an example.

OR

- b) Convert the $(c * b) - (a + f - x)/(2 * d)$ expression to postfix and prefix.

- 3 a) Demonstrate algorithm with suitable example and also discuss analysis of the algorithms.

OR

- b) Explain how infix expressions are converted to polish notation. Illustrate your answer with suitable example.

- 4 a) Discuss about Queues. Write algorithm for inserting and deleting an element in to Queue using linked List

OR

- b) Define Double linked list. Write the pseudo code to insert and delete element from beginning of Double linked list

- 5 a) Discuss about In order, Pre order, Post orders with suitable examples and write Pseudo code for the tree traversal.

OR

- b) Write the quick sort algorithm and trace the quick sort algorithm with your own example.

- 6 a) Discuss in detail the Graph representation mechanisms and write the algorithm for Breadth first search

OR

- b) Define Collision? Explain about Collision resolution techniques with examples

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FACULTY OF SCIENCE
B.Sc (VI – Semester) Examination
COMPUTER SCIENCE
Paper VI (New)
Web Technologies

Time: 3 Hours]

[Max Marks: 80

Section A – (Marks: $8 \times 4 = 32$)

1. Answer any eight questions in short form:

- a) Define XHTML.
- b) Write the syntax for inserting an image in HTML.
- c) What are meta elements in XHTML? ✓
- d) Explain inline styles in CSS. ✓
- e) What is the purpose of the 'switch' statement in JavaScript? ✓
- f) Differentiate between 'while' and 'do...while' loops. ✓
- g) What is an array in JavaScript? ✓
- h) Explain the 'onmouseover' event with an example. ✓
- i) What is the use of the Math object in JavaScript? ✓
- j) Define XML namespaces.
- k) What is a Document Type Definition (DTD)? ✓
- l) Explain the role of XMLHttpRequest in Ajax.

Section B – (Marks: $4 \times 12 = 48$)

Answer all questions

- 2. a) Explain the structure of an XHTML document with examples.
OR
b) Discuss the CSS box model with neat diagrams.
 - 3. a) Write a JavaScript program to find the factorial of a number using recursion.
OR
b) Explain different types of control structures in JavaScript with examples. ✓
 - 4. a) Explain DOM and its role in web development. ✓
OR
b) Discuss JavaScript objects: String, Date, and Boolean with examples.
 - 5. a) Explain XML Schema and its advantages over DTD. ✗
OR
b) Write notes on Ajax-enabled Rich Internet Applications with an example. ✓
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FACULTY OF SCIENCE
B.Sc. (IV Semester) Examination
SKILL ENHANCEMENT COURSE-III
(Fundamentals of Python)
(New)

660243012

Time: 2 Hours]

[Max. Marks: 40

Section A – (Marks: $4 \times 4 = 16$)
(Short Answer Questions)

1. Answer any **four** of the following:
 - (a) Write about the history and features of Python.
 - (b) Discuss the various operators available in Python.
 - (c) Write about for loop.
 - (d) Basic tuple operations.
 - (e) Function.
 - (f) File.

Section B – (Marks: $2 \times 12 = 24$)
(Essay Type Questions)

Answer the following questions.

2. (a) Explain in detail the list and tuples methods with suitable examples.
Or
(b) Discuss the various string operations in python with suitable examples.
 3. (a) Explain the different basic string operations in Python.
Or
(b) Discuss in detail about passing arguments to a function.
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FACULTIES OF ARTS / SCIENCE / COMMERCE / SOCIAL SCIENCES

B.A. / B.A.(L) / B.Sc. / B.Com. / B.B.A. (I Semester) Examination

FUNDAMENTALS OF COMPUTERS

Paper I

(AEC-1)

Time: 2 Hours]

[Max. Marks: 40

Section A – (Marks: $4 \times 4 = 16$)

1. Answer any **four** of the following:

- (a) Define Computers and applications
- (b) Characteristics of Computers
- (c) Input Devices
- (d) Logical Gates
- (e) BCD code
- (f) System Software.

Section B – (Marks: $2 \times 12 = 24$)

Answer all questions.

2. (a) Explain in detail about the classifications of Computers.

Or

(b) Discuss about Storage Devices.

3. (a) Explain about Boolean algebra and Venn diagrams.

Or

(b) Discuss in detail about Firmware and Middle ware with examples.

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FACULTY OF COMMERCE AND BUSINESS MANAGEMENT

B.Com. (V Semester) Examination

E-COMMERCE

Paper III (b)

(New)

Time: 3 Hours]

[Max. Marks: 50

Section A – (Marks: $5 \times 3 = 15$)

Answer all short questions in serial order.

1. Answer any five of the following questions.
 - (a) List the advantages and disadvantages of E-Commerce.
 - (b) Explain the applications of E-Commerce.
 - (c) Explain the types of networks.
 - (d) Write in brief about E-Banking.
 - (e) Write brief advantages of EDI.
 - (f) Write about B2C and B2B models.
 - (g) What is T-1 line?
 - (h) Explain pros & cons of online shopping.

Section B – (Marks: $5 \times 7 = 35$)

Answer all essay type questions.

2. (a) Explain the main activities of E-Commerce.
Or
(b) Write in detail about the technical components of E-Commerce.
3. (a) Write a brief note on evolution of internet names and internet organisations.
Or
(b) Explain in detail about the Web promotion activity.
4. (a) Briefly explain the limitations and applications of EDI.
Or
(b) Write in detail about an Electronic Data Interchange (EDI) model.
5. (a) Explain about the concept of Personalization of E-Commerce.
Or
(b) Write a detailed note on different types of Electronic Payment Systems.
6. (a) What is Search Engine? Write about its optimization.
Or
(b) Write a short note on the role of Digital Marketing.

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FACULTY OF SCIENCE
B. Sc (V- Semester) Examination
COMPUTER SCIENCE
Paper-V (a) (New)
Programming in Java

Time: 3 Hours]

[Max.Marks:80

Section-A (Marks: 8 x 4=32)

1. Answer any ~~Eight~~ of the following:

- a) Differentiate between class and object.
- b) Discuss Java data types, variables, type casting, and operators with suitable examples.
- ☒ c) Write short notes on conditional statements in Java.
- d) What are the this and static keywords?
- ☒ e) Explain inner classes with examples.
- f) Explain method overloading and constructor overloading in Java with programs.
- g) How do you create a thread using the Runnable interface?
- h) What are userdefined exceptions?
- ☒ i) Explain multithreading in Java.
- ☒ j) What is an applet? Explain the life cycle of an applet.
- k) Explain JFrame and JPanel with examples.
- ☒ l) Write notes on layout managers.

Section-B (Marks: 4x12=48)

Answer all questions

- 2 a). Explain the features of Java in detail and describe how Java achieves platform independence
Or
b). Describe the structure of a Java program with a complete example.
 - 3 a). Explain abstract classes, interfaces, and differences between them with examples.
Or
b). Discuss one-dimensional and two-dimensional arrays in Java with examples.
 - 4 a). What is exception handling? Explain types of exceptions and handling techniques with examples
Or
b). Explain Java I/O streams in detail.
 - 5 a). Discuss Swing components used for GUI development
Or
b). Create a Student registration form using AWT components
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